

# Soo Min Kwon

✉ soominkwon0402@gmail.com | 🎓 Google Scholar | 🌐 soominkwon | 🌐 soominkwon.github.io

## EDUCATION

### University of Michigan

*Ph.D., Electrical Engineering and Computer Science*

Sept. 2022 – May 2026

Ann Arbor, MI

- **Thesis:** “Deep Learning through Low-Dimensional Representations: Theory and Algorithms”
- **Advisor(s):** Prof. Laura Balzano and Prof. Qing Qu

### Rutgers University

*M.S., B.S., Electrical and Computer Engineering*

Sept. 2016 – May 2022

New Brunswick, NJ

- **Thesis:** “Optimization Problems with Low-Dimensional Tensor Structures”
- **Advisor:** Prof. Anand D. Sarwate

## EXPERIENCE

### Google

*Student Researcher*

July 2025 – Nov. 2025

New York, NY

- Hosted by Himanshu Jain and Ziteng Sun
- Developed CoDistill-GRPO, a novel variant of GRPO aimed at improving small language model performance via co-distillation training with larger language models
- Tested the effectiveness of CoDistill-GRPO on the Qwen and Llama model families and with models up to 8B parameters

### Amazon

*Applied Scientist Intern*

July 2024 – Nov. 2024

Seattle, WA

- Developed a causal inference framework using deep learning methods for the SCOT team that reduced variance estimates of average treatment effects by over 10%

### LinkedIn

*Applied Research Scientist Intern*

May 2022 – Aug. 2022

Sunnyvale, CA

- Productionized a machine learning pipeline for the infrastructure team, reducing MAPE by over 15% in forecasting hardware needs for the next calendar year

## SELECTED PUBLICATIONS

- [1] **Out-of-Distribution Generalization of In-Context Learning: A Low-Dimensional Subspace Perspective** [Paper]  
*S. M. Kwon<sup>†</sup>, A. S. Xu<sup>†</sup>, C. Yaras, L. Balzano, Q. Qu*  
AISTATS 2026; AISTATS 2026 CauScale Workshop (**Oral**); DeepMath 2025 (**Oral**)
- [2] **An Overview of Low-Rank Structures in the Training and Adaptation of Large Models** [Paper]  
*L. Balzano, T. Ding, B. D. Haeffele, S. M. Kwon, Q. Qu, P. Wang, Z. Wang, C. Yaras ( $\alpha$ - $\beta$  Order)*  
IEEE Signal Processing Magazine
- [3] **Learning Dynamics of Deep Matrix Factorization Beyond the Edge of Stability** [Paper]  
*A. Ghosh<sup>†</sup>, S. M. Kwon<sup>†</sup>, R. Wang, S. Ravishankar, Q. Qu*  
ICLR 2025
- [4] **BLAST: Block-Level Adaptive Structured Matrices for Efficient DNN Inference** [Paper]  
*C. Lee, S. M. Kwon, Q. Qu, H. Lee*  
NeurIPS 2024

- [5] **On the Relationship Between Small Initialization and Flatness in Deep Networks** [Paper]  
*S. M. Kwon, L. Ding, L. Balzano, Q. Qu*  
 ICLR 2024 Workshop on Bridging the Gap Between Practice and Theory in Deep Learning
- [6] **Efficient Compression of Overparameterized Deep Models through Low-Dimensional Learning Dynamics** [Paper]  
*S. M. Kwon, Z. Zhang, D. Song, L. Balzano, Q. Qu*  
 AISTATS 2024
- [7] **Solving Inverse Problems with Latent Diffusion Models via Hard Data Consistency** [Paper]  
*B. Song<sup>†</sup>, S. M. Kwon<sup>†</sup>, Z. Zhang, X. Hu, Q. Qu, L. Shen*  
 ICLR 2024 (**Spotlight, Top 5%**)

## PREPRINTS

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- [1] **CoDistill-GRPO: A Co-Distillation Recipe for Efficient Group Relative Policy Optimization** [Paper]  
*S. M. Kwon, Himanshu Jain, Ziteng Sun, Ananda Theertha Suresh, Sanjiv Kumar*
- [2] **The Effect of Training Task Diversity on In-Context Learning through the Lens of Low-Dimensional Subspaces** [Paper]  
*S. M. Kwon, Alec S. Xu, Can Yaras, Dogyoon Song, Laura Balzano, Qing Qu*
- [3] **Dynamic Subspace Estimation from Undersampled Data using Grassmannian Geodesics** [Paper]  
*S. M. Kwon<sup>†</sup>, C. Blocker<sup>†</sup>, H. Raja, J. Fessler, L. Balzano*
- [4] **Decoupled Data Consistency for Solving General Inverse Problems with Diffusion Models** [Paper]  
*X. Li, S. M. Kwon, I. Alkhouri, S. Ravishankar, Q. Qu*

## AWARDS & HONORS

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|                                                     |             |
|-----------------------------------------------------|-------------|
| Harvey G. and Joyce H. Behner Graduate Fellowship   | 2024        |
| University of Michigan PhD Rackham Merit Fellowship | 2023        |
| Rutgers ECE Outstanding Master's Student Award      | 2022        |
| Rutgers ECE Outstanding Teaching Assistant Award    | 2021        |
| Rutgers ECE Departmental Leadership & Service Award | 2020        |
| Rutgers WINLAB GA/TA Grant                          | 2020        |
| Rutgers University Dean's List                      | 2018 – 2020 |

## INVITED TALKS

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|------------------------------------------------|------|
| 2025 INFORMS Annual Meeting                    | 2025 |
| Math Machine Learning Seminar @ MPI MIS + UCLA | 2025 |